

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

1. Consider the function $y = \begin{cases} x^2 & \text{if } x < 2 \\ x & \text{if } x \geq 2 \end{cases}$ on the interval $[0, 4]$.

a) Estimate the area under the curve on $[0, 4]$ using a left endpoint Reimann sum with 4 rectangles.

b) Estimate the area under the curve on $[0, 4]$ using a right endpoint Reimann sum with 2 rectangles.

c) Estimate the area under the curve on $[0, 4]$ using a midpoint Reimann sum with 2 rectangles.

2. Suppose that the speed of a runner was always increasing for $10 \leq t \leq 30$. Use the data on the runner's speed to give an upper and lower estimate on the distance travelled from 10 seconds to 30 seconds.

time (seconds)	10	12	20	24	30
speed (ft/sec)	7	8	9	9.5	10

3. Consider the function $f(x) = x^2 - 4x + 5$ over the interval $[0, 2]$.
- a) Show that $f(x)$ is decreasing over $(0, 2)$.

b) Suppose we attempt to estimate the area between $f(x)$ and the x -axis on $[0, 2]$. Will the sum created by using a Left Reimann sum (L_n) be an over or underestimate? Will the sum created by using a Right Reimann sum (R_n) be an over or underestimate? Explain.

c) Suppose you split $[0, 2]$ into n subintervals of equal length ($\Delta x = \frac{2}{n}$). Graphically demonstrate how the error from L_n or R_n is less than $[f(0) - f(2)]\Delta x = \frac{8}{n}$.

- d) Use what you showed in part (c) to explain what happens to the error from either L_n or R_n as we divide the interval into more subintervals? How do L_n and R_n compare as $n \rightarrow \infty$?

4. Consider the function $f(x) = \sqrt{x-1}$.

- a) Estimate the area under this curve from $[1,10]$ using right endpoints with $n = 3$.

- b) Create a formula to estimate the area under the curve with n rectangles.

- c) What could you add to the formula above for the formula to represent the total area of the region?